(a)

In [7]:
```
import numpy as np

t = np.array([0, 1, 2, 3, 4, 5])
x = np.array([2.0, 3.5, 5.0, 4.0, 2.0, 1.0])

velocity = []
for i in range(1, len(t) - 1):
    v = (x[i + 1] - x[i - 1]) / (2 * (t[i + 1] - t[i - 1]))
    velocity.append(v)

for i in range(len(velocity)):
    print(f"Velocity at t = {t[i + 1]} s: {velocity[i]:.2f} m/s")
```

```
Velocity at t = 1 s: 0.75 m/s
Velocity at t = 2 s: 0.12 m/s
Velocity at t = 3 s: -0.75 m/s
Velocity at t = 4 s: -0.75 m/s
```

(b)

In [6]:
```
import numpy as np

t = np.array([1, 2, 3, 4])
v = np.array([1.5, 0.25, -1.5, -1.5])

acceleration = []
for i in range(1, len(t) - 1):
    a = (v[i + 1] - v[i - 1]) / (2 * (t[i + 1] - t[i - 1]))
    acceleration.append(a)

for i in range(len(acceleration)):
    print(f"Acceleration at t = {t[i + 1]} s: {acceleration[i]:.2f} m/s^2")
```

```
Acceleration at t = 2 s: -0.75 m/s^2
Acceleration at t = 3 s: -0.44 m/s^2
```

In [ ]:
```

```